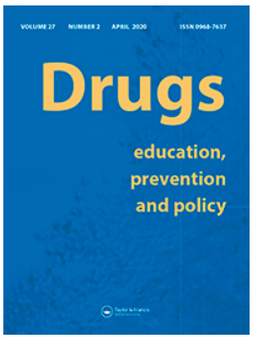

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An economic analysis of the demand for cannabis: some results from South Africa

Shaun Riley, Nicole Vellios and Corné van Walbeek

School of Economics, University of Cape Town, Cape Town, South Africa

ABSTRACT

Background: The full legalization of cannabis is currently being deliberated in South Africa. We aim to better understand how the market operates by investigating how cannabis consumers respond to price changes and what factors are associated with the price that consumers pay.

Methods: We estimate the conditional price elasticity of demand using data on price, quantity, and quality from a cross-sectional survey of cannabis consumers across South Africa. We also investigate to what extent quality, quantity bought, income, and demographic variables influence the price paid for cannabis.

Results: The price of cannabis differs greatly by quality: medium-quality is almost double the price of low quality, while the price of high-quality cannabis is nine times that of low-quality cannabis. Depending on the model specification, we estimate that a 10% increase in the price per gram of cannabis is associated with an approximate 5–6% decrease in the quantity demanded. There is some evidence that the demand for medium- and high-quality cannabis is more price elastic than low quality. The price per gram is negatively associated with the quantity purchased and positively associated with income.

Conclusions: Price is a significant determinant of cannabis consumption. Increases in prices result in a decrease in quantity demanded.

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Cannabis; marijuana; dagga; price elasticity; South Africa

Introduction

In recent years, the legality of cannabis production, distribution, and use in South Africa has been at the forefront of public debate. Two landmark court cases took place in 2017 in the provincial divisions of the High Court of South Africa (Western Cape and Gauteng), which have general jurisdiction over their defined areas.

In the first case, the Western Cape High Court (WCHC) recommended on 31 March 2017 that the possession, cultivation, and use of cannabis at home for private use by adults be decriminalized (High Court of South Africa, 31 March 2017). The ruling found that the policing of cannabis under the auspices of the Drugs and Drugs Trafficking Act (140 of 1992) often led to the infringement of individual rights, such as privacy. The recommendation did not, however, support the complete legalization of cannabis, but rather recognized the harms generated by its regulation and therefore called for the decriminalization of related offenses should they meet defined prerequisites (High Court of South Africa, 31 March 2017). The WCHC ruled that Parliament must change sections of the Drug Trafficking Act, as well as the Medicines Control Act, within 24 months (High Court of South Africa, 31 March 2017). The state, which includes the Ministries of Health, Police, and Social Development (among others), appealed the WCHC ruling (Constitutional Court of South Africa, 13 September 2017).

On 18 September 2018, the Constitutional Court of South Africa denied the appeal and ruled that South Africans can now smoke cannabis in the privacy of their own homes (Constitutional Court of South Africa, 18 September 2018).

In the second case, the 'dagga couple' – Myrtle Clarke and Julian Strobbs – defended their right to use cannabis at the Northern Gauteng High Court (NGHC), in a case termed the 'Trial of the Plant' (news24, 6 November 2017). The case was heard from 31 July 2017 to 18 August 2017, when arguments for and against the legalized regulation of dagga were presented (news24, 6 November 2017). The trial has yet to be completed, having run out of the allotted time in 2017. This case followed a series of other court appearances resulting from the arrest of the couple for using and dealing cannabisin in 2010.

The legalization of cannabis is currently receiving global attention as many countries are considering this policy. In June 2018, Canada passed a bill to legalize cannabis, making Canada the second country (after Uruguay) to legalize

CONTACT Nicole Vellios nicole.vellios@uct.ac.za School of Economics, University of Cape Town, Middle Campus, Rondebosch, Cape Town 7700, South Africa

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cannabis for recreational use (Sapra, 2018; Watson & Erickson, 2018). By June 2018, nine states and the District of Columbia in the US allowed recreational cannabis use, while 30 states allowed cannabis for medical use (Sapra, 2018). After legalizing cannabis in 2014, Colorado started collecting a substantial amount of tax from cannabis sales: in 2015, $113 million was collected in retail cannabis tax revenue – twice the amount collected from alcohol taxes (Scarboro & Bishop-Henchman, 2016).

If cannabis is fully legalized in South Africa, there are two possible price effects. First, if the government imposes excise taxes on cannabis, prices will increase (and government will generate revenue). Second, the price of cannabis is likely to decrease since legalization would result in eliminating the risk associated with buying and selling cannabis. A 2013 paper that investigates the challenges of conducting a cost-benefit analysis of cannabis policies reports that data is insufficient to determine the shape of the demand curve for cannabis, or the shift of the demand curve should legislation occur (Shanahan & Ritter, 2013). A related question is whether alcohol and cannabis are substitutes or complements, and whether this varies by age, gender, and use patterns (Shanahan & Ritter, 2013).

To evaluate the potential effects that legalization and the imposition of an excise tax may have, an understanding of the price elasticity of demand is helpful. For example, the demand for cigarettes in South Africa responds to changes in the price (with an estimated price elasticity of demand of between −0.5 and −0.7). Increased excise taxation curbs the demand for cigarettes for current consumers, reduces initiation among youth, and raises revenue for government (Van Walbeek, 2015; Vellios & Van Walbeek, 2016).

In this article, we estimate the conditional price elasticity of demand for cannabis using data on price, quantity, and quality from a cross-section of cannabis consumers. We focus on the *conditional* price elasticity of demand, which is the percentage change in consumption that corresponds to a 1% change in price among individuals who use cannabis. The *participation* elasticity of demand, which is estimated in most papers (Gallet, 2014), denotes the percentage change in the number of individuals who report any substance use in response to a 1% change in price. The sum of the conditional and the participation elasticities is the total price elasticity of demand, which represents the percentage change in total consumption corresponding to a 1% change in price.

Two recent literature reviews, both published in 2014, investigate the elasticity of demand for cannabis (Gallet, 2014; Pacula & Lundberg, 2014). Pacula and Lundberg find that there is a sizable literature that analyzes the impact of price on cannabis use prevalence (i.e. the participation elasticity), but very little research that considers the conditional price elasticity (Pacula & Lundberg, 2014). In a meta-analysis of 14 studies (mostly from the US and Australia, and one from India) that considers various age groups, Gallet finds that the participation elasticity of demand for cannabis lies between −0.15 and −0.31.

The few studies that consider the conditional price elasticity of demand for cannabis find that the estimates lie in the inelastic range, and are typically smaller than 0.6 in absolute terms. For example, Kilmer, Caulkins, Pacula, MacCoun, & Reuter (2010) estimate a conditional elasticity of −0.24 in the US. Ouellet, Macdonald, Bouchard, Morselli, & Frank (2017), using crowd-sourced transaction data (which we also use in this study), find that the conditional price elasticity in Canada lies between −0.42 and −0.60.

In terms of the prevalence of cannabis use in South Africa, estimates vary substantially. A 2002 survey found past-month prevalence rates of 9.1% (13.7% male, 5.5% female) among South African youths (13–19 years) (Reddy et al., 2003). A 2005 survey found that 2.1% of adults (aged 15 and older) use cannabis on a daily basis (Shisana et al., 2005). These percentages likely understate actual prevalence as people tend to under-report socially undesirable behaviors (Pick et al., 2003). For example, Pick et al. (2003) found that self-confessed past-month use among South African mine workers was 2.3%, while a urine analysis of these respondents indicated that 9.1% of workers had cannabinoids in their system (Pick et al., 2003).

A 2018 United Nations publication reports that while most of the cannabis produced in Africa is for consumption within the region, a number of African countries (South Africa, Nigeria, Ghana, and Zambia) have identified European countries as the final destination, notably the United Kingdom and the Netherlands (United Nations Office on Drugs and Crime, 2018). A 2014 study analyzed cannabis confiscated by the South African Police Service's Forensic Laboratories to assess the quality with regard to potency of the cannabis in a localized South African market (Londt, 2014). The study found that cannaboid content in cannabis has increased in the past few decades (Londt, 2014).

This article focuses on the effect of price on cannabis consumption. We do not investigate other non-monetary aspects that may affect a person's decision to smoke cannabis (such as effective police enforcement, religious beliefs, and the use of other substances). To the extent that these non-monetary aspects increase the effective cost of using cannabis, the price is understated. Among adolescents and young people, other predictors (which we do not explore) include high levels of delinquency, lower academic performance, drug use among friends, and stressful life events (Windle & Wiesner, 2004).

To our knowledge, this is the first study in South Africa that empirically estimates the conditional price elasticity of cannabis demand using actual transaction histories reported directly from consumers.

Methods

Data were collected in August and September 2017 from a non-random sample of cannabis consumers in South Africa. Respondents were recruited primarily through social media, and through contacts with pro-cannabis organizations. Twitter and Facebook pages were created to inform potential respondents of the research. Periodic 'posts' and 'tweets' were sent out as reminders during data collection. Respondents anonymously answered a 20-question survey covering demographics, income, frequency of use, and information about

their most recent cannabis transactions: amount purchased, price of purchase, and quality.

Respondents were asked to record the details of their purchase with a clarifying note: 'this is referring only to unprocessed forms of cannabis, i.e. no hash, oil, wax, etc.).' In South Africa, it is common for low-quality cannabis to include many seeds, medium-quality to be with few-to-no seeds (sensimilla), and high-quality to be without seeds and usually grown indoor.

No incentives were offered to participate. Ethics approval was granted by the University of Cape Town.

Of the 2,250 responses submitted, 1,977 were used for the data analysis. We dropped nine observations where informed consent was not given, 67 observations with missing information for two crucial variables (quality and price) and 166 observations where monthly purchases were reported as more than 200 g. We suspect that people who purchase more than 200 g per month were trading in cannabis because a consumer would need to smoke about 20 joints a day to consume 200 g of cannabis a month. To eliminate the impact of outliers on the price, we deleted a further 31 observations (namely the first and last percentile of the price per gram of low-, medium-, and high-quality cannabis).

We estimate two different models, represented by Equations (1) and (2) below. The first model estimates the conditional price elasticity of demand, together with other explanatory variables. We estimate the demand for cannabis in a number of different specifications, which is generically presented as follows:

$$LnQ_i = \beta_1 + \beta_2 LnP_i + \beta_3 X_i + \varepsilon_i \quad (1)$$

where Q_i is monthly consumption (in grams) for consumer i, P_i is the reported price per gram, and X_i is a vector of personal characteristics. The parameter of interest is β_2, since this indicates by what percentage consumption of cannabis changes in response to a 1% change in the price of the product.

The second model estimates the covariates of the price that buyers of cannabis pay for the product. We appreciate that Equation (2) raises issues of endogeneity, since the covariate in Equation (1) becomes the dependent variable in Equation (2). We address this issue as a limitation in the discussion section, but note at this point that a similar study, also using transaction data, found that the degree of endogeneity is limited and has not greatly affected the values of the coefficients (Davis, Geisler, & Nichols, 2016). As will be shown in the results section, this specification provides very useful insights about the functioning of the cannabis market. Note that in Equation (1) *monthly quantity demanded* is used, whereas in Equation (2) *quantity purchased* at the most recent transaction is used.

$$LnP_i = \beta_1 + \beta_2 LnQ_i + \beta_3 LnI_i + \beta_4 X_i + \varepsilon_i \quad (2)$$

where P_i is the reported price per gram by consumer i, Q_i is quantity purchased (in grams), I_i is income and X_i is a vector of personal characteristics. The data analysis was conducted using Stata version 14 (StataCorp, College Station, TX).

Results

Of those sampled, almost three-quarters are male (Table 1). The majority of the sample is comprised of Whites (74.6%), followed by Coloreds (a non-derogatory term in South Africa, referring to mixed race descendants of indigenous, East Asian and European populations, 12.0%), Africans (9.0%), and Asians/Indians (4.5%). South Africans were officially classified into these population groups under apartheid and even though these classifications were abolished in the 1990s, many people still think of themselves in these terms. The sample is also skewed toward those residing in the two richest provinces, the Western Cape (47.7%) and Gauteng (30.1%). Income among respondents is distributed widely. Two-thirds of respondents use cannabis daily, while the remainder uses it more occasionally. More than half of consumers have been using cannabis for more than 5 years.

The median (mean) price per gram differs greatly by quality: low (R8.33; R11.17) ($0.63; $0.84), medium (R18.75; R22.96) ($1.41; $1.73) and high (R93.54; R88.37) ($7.03; $6.64).[1] The standard deviations for all three-price categories are large: R8.45 ($0.64) for low quality, R15.74 ($1.18) for medium quality and R54.53 ($4.1) for high quality, indicating a wide variance in prices. The category with the most observations is medium-quality ($N = 1066$), followed by high quality ($N = 754$), and low quality ($N = 157$).

Table 1. Sample characteristics.

Variable	Proportion (%)	*N*
Gender		
Male	72.8	1428
Female	27.2	533
Population group		
African	9.0	168
Colored	12.0	223
Asian/Indian	4.5	83
White	74.6	1389
Age group		
<20	9.0	178
20–29	66.5	1308
30–39	16.5	324
40–49	4.7	92
50+	3.4	66
Province		
Western Cape	47.7	943
Gauteng	30.1	594
Other	22.3	440
Monthly income		
<R1000 (<$75)	9.7	192
R1000–R4999 ($75–$376)	27.6	546
R5000–R9999 ($376–$752)	20.8	411
R10 000–14 999 ($752–$1128)	14.4	284
R15 000–R19 999 ($1128–$1504)	8.2	163
R20 000–24 999 ($1504–$1880)	6.5	129
≥R25 000 (≥$1880)	12.7	252
Intensity		
Once per month (or less)	5.0	97
2–3 times per month	4.3	85
Once per week	5.2	102
2–3 times per week	19.5	381
Daily	66.1	1294
Years as cannabis consumer		
1–5 years	44.1	871
6–10 years	32.6	644
≥11 years	23.4	462

Where the sum per category does not equal 1977, this is due to missing values. Monthly income was converted to USD using an average exchange rate for 2017 ($1 = R13.30).

Table 2. Monthly quantity demanded by quality and gender (%).

	Low		Medium		High	
	Male	Female	Male	Female	Male	Female
0–10 g	7.3	6.8	10.4	14.6	26.0	31.8
10.1–20 g	15.4	18.2	15.1	14.9	19.9	21.2
20.1–30 g	17.3	13.6	20.7	19.2	19.5	15.6
30.1–50 g	17.3	25.0	21.0	24.7	12.3	11.2
50.1–100 g	24.6	15.9	21.9	18.2	14.8	14.5
>100 g	18.2	20.4	10.9	8.4	7.6	5.6
Total (%)	100	100	100	100	100	100
Total (*n*)	110	44	743	308	569	179

An average exchange rate for 2017 ($1 = R13.30) is used.

Table 3. Price per gram by quality and gender (%).

	Low		Medium		High	
	Male	Female	Male	Female	Male	Female
R0–R10 ($0-$0.75)	66.4	62.2	25.8	25.9	3.5	6.2
R10.1–R20 ($0.76–$1.50)	26.4	20.0	35.1	35.0	8.6	7.3
R20.1–R30 ($1.51–$2.26)	6.4	11.1	17.4	17.2	9.6	11.7
R30.1–R50 ($2.26–$3.76)	0.9	6.7	16.5	17.8	12.8	14.5
R50.1–R100 ($3.77–$7.52)	–	–	5.2	4.2	30.0	25.1
>R100 ($7.52)	–	–	–	–	35.6	35.2
Total (%)	100	100	100	100	100	100
Total (*n*)	110	45	747	309	571	179

An average exchange rate for 2017 ($1 = R13.30) is used.

Males and females who purchase high-quality cannabis tend to buy smaller quantities compared to buyers of low quality (Table 2). For example, 26.0% of males purchase 0–10 g of high-quality cannabis (females 31.8%), while only 7.3% of males purchase 1–10 g of low-quality cannabis (females 6.8%). Whereas large purchases (>100 g) of low-quality cannabis are bought by approximately a fifth of male and females, only 7.6% of males and 5.6% of females purchase >100 g of high-quality cannabis.

Two-thirds of males and females purchase low-quality cannabis for R10 ($0.75) or less per gram, while about a quarter of males and females spend this amount on medium quality cannabis (Table 3). Very few males (3.5%) and females (6.2%) spend R10 ($0.75) or less per gram on high-quality cannabis. No consumers who use low-quality cannabis spend more than R50 ($3.76) per gram on their purchase. No consumers who use medium quality cannabis sped more than R100 ($7.52) per gram. More than a third of male and female consumers of high-quality cannabis spend more than R100 ($7.52) per gram.

In Table 4, we present the estimated demand equations for cannabis as described in Equation (1) in the methods section. Columns 1, 2a, 3a, 4a, and 5a present the regression results while columns 2b, 3b, 4b, and 5b present the coefficients (which are associated with dummy variables) converted to percentages using the formula $(e^{\beta_i} - 1)\times 100$ (Gujurati, 2003).

Considering the simplest possible specification, where the log of monthly quantity is regressed on the log of price per gram, we see that a 1% increase in the price per gram of cannabis is associated with a 0.501% (95% CI: 0.461%; 0.540%) decrease in consumption (Table 4, Column 1). This price elasticity estimate is in the inelastic range and indicates that a given percentage change in the price is associated with a less than proportional change in consumption. The positive coefficient on the quality dummy for medium- and high-quality dummy variables (Table 4, Column 2a) indicates that *if the price is the same*, people would increase the demand for medium-quality by 35.5% (95% CI: 17.1%; 56.7%) and for high-quality cannabis by 81.3% (95% CI: 52.5%; 115.3%). The specification in Table 4 Columns 2a and 2b is presented graphically in Figure 1(a). The demand curve shifts outwards as quality improves. Figure 1(b) corresponds to the regression in Table 4, Columns 3a and 3b. Even though the demand curve for low-quality cannabis looks contradictory (in that for very high prices, the quantity demanded of low-quality cannabis is greater than that of medium- or high-quality cannabis), this does not have any practical effect. For these high prices (and correspondingly low quantities), the lines are extrapolations of the relationship that is concentrated between the horizontal dashed lines. In fact, the dashed lines indicate that 98% of actual prices fall within the range where this contradiction does not occur. Figure 1(b) and Table 4 Column 3a indicates that, if one allows the price elasticities to vary by perceived quality, the demand for medium- and high-quality cannabis is somewhat more price elastic than the demand for low-quality cannabis.

Unsurprisingly, users who use cannabis daily demand more of the product than users who use cannabis less often (Table 4, Column 4a). Daily users demand 125.5% (95% CI: 107.6%; 145.1%) more per month than non-daily users. Column 4a explains about 40% of the variation in the log of monthly demand for cannabis.

In order to account for any other covariates of the demand for cannabis we add gender, population group, age, province, and log of income in column 5a of Table 4. These additional variables do not add much explanatory power to the regression equation. Nevertheless, a number of the additional variables are statistically significant. Females who purchase cannabis demand 11.2% (95% CI: 3.3%; 18.5%) less per month than males. While there are no significant population group effects, we kept this variable in the model as it is generally an important covariate in South Africa. Consumers aged 30–39 demand 16.6% (95% CI: 0.2%; 30.3%) less than those aged less than 20. Compared to the Western Cape, cannabis users in Gauteng and other provinces demand 12.2% (95% CI: 2.3%; 23.0%) and 11.3% (95% CI: 0.8%; 22.9%) more cannabis, respectively. Income is not a significant determinant of the demand for cannabis.

In Table 5, we consider factors that influence the price of cannabis using Equation (2). The dependent variable is the natural logarithm of the price per gram.

As indicated earlier, there are substantial price differences across the three quality categories. There are also very large socio-economic differences between the different population groups. In order to determine whether the price paid by the different population groups varies across the three quality categories, we interacted quality with population group. The base category is Whites who consume low-quality cannabis. Africans pay 53.8% less than Whites for low-quality cannabis (95% CI: −66.5%; −36.4%). Coloreds (−31.9%; 95% CI: −46.6%; −13.0%) and Indians (−23.2%; 95% CI: −43.1%; 3.6%) also pay less than Whites, but more than Africans, for

Table 4. OLS estimates of cannabis demand.

	(1)	(2a)	% (2b)	(3a)	% (3b)	(4a)	% (4b)	(5a)	% (5b)
Ln (price per gram)	−0.501***	−0.613***		−0.462***		−0.528***		−0.493***	
	(0.020)	(0.024)		(0.071)		(0.071)		(0.073)	
Quality (base: low)									
Medium		0.304***	35.5	0.584***	79.3	0.201	22.3	0.353*	42.3
		(0.074)		(0.192)		(0.189)		(0.198)	
High		0.595***	81.3	1.095***	198.9	0.634***	88.5	0.646***	90.8
		(0.088)		(0.218)		(0.212)		(0.220)	
Ln(price per gram)*quality (base: low)									
Ln(price per gram)*medium				−0.138*		−0.026		−0.078	
				(0.079)		(0.078)		(0.081)	
Ln(price per gram)*high				−0.194**		−0.070		−0.086	
				(0.080)		(0.078)		(0.080)	
Intensity (base: less than daily)									
Daily						0.813***	125.5	0.815***	125.9
						(0.042)		(0.043)	
Gender (base: male)									
Female								−0.119***	−11.2
								(0.044)	
Population group (base: White)									
African								0.048	4.9
								(0.073)	
Colored								0.075	7.8
								(0.059)	
Asian/Indian								0.028	2.8
								(0.093)	
Age (base: <20)									
20–29								−0.095	−9.1
								(0.075)	
30 − 39								−0.182**	−16.6
								(0.092)	
40 − 49								−0.105	−10.0
								(0.117)	
50+								−0.149	−13.8
								(0.134)	
Province (base: Western Cape)									
Gauteng								0.115**	12.2
								(0.047)	
Other								0.107**	11.3
								(0.051)	
Ln (income)								0.024	
								(0.022)	
Constant	5.005***	4.991***		4.671***		4.366***		4.117***	
	(0.065)	(0.085)		(0.163)		(0.166)		(0.250)	
Observations	1969	1969		1969		1952		1824	
R-squared	0.254	0.271		0.273		0.406		0.415	

Dependent variable: logged quantity of cannabis purchased monthly (grams). Robust standard errors in parentheses.
***$p<.01$, **$p<.05$, *$p<.1$.

low-quality cannabis. As expected, all population groups pay a higher price for medium-quality and high-quality cannabis than for low-quality cannabis. Within the medium-quality category (as for the low-quality category), Whites pay the highest price per gram, followed by Indians, Coloreds, and Africans, in that order. For high-quality cannabis, Indians report paying the highest price, followed by Whites, Coloreds, and Africans.

Both quantity purchased and income are statistically significant covariates of the price that cannabis users report. A 1% increase in purchased quantity is associated with a 0.436% decrease in the price paid (95% CI: −0.467%; −0.406%). This implies that drug dealers give their customers a substantial 'volume discount.' Also, richer users pay a higher price for the product than poorer users – a 1% increase in income is associated with a 0.103% increase in the price paid (95% CI: 0.071%; 0.135%). This income effect is over and above the racial effect, where the richer population groups (i.e. Whites and Indians) pay substantially more than the poorer population groups (i.e. Africans, and to a lesser extent, Coloreds).

The price paid by females is 5.8% less than the price paid by males (95% CI: −11.8%; −0.5%). Young cannabis users (age < 20) report paying a lower price (−13.2%; 95% CI: −21.6%; −3.7%) than users aged 20–29. Older users (age 50+) also report a substantially lower price than users in the 20–29 age group (−16.9%; 95% CI: −29.9%; −1.5%).

Cannabis users in Gauteng (the wealthiest province in the country) pay a marginally higher price (6%; 95% CI: −0.9%; 13.3%) than users who live in the Western Cape. Daily users also pay a marginally higher price (7.0%; 95% CI: 0.7%; 13.8%) than users who consume cannabis less than daily.

Discussion

While there are many studies from a variety of countries that look at participation elasticities of cannabis use, few focus

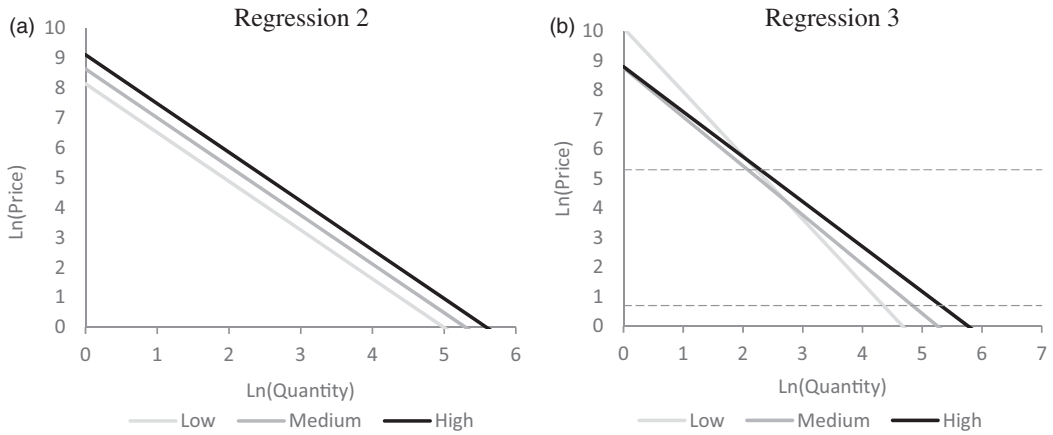

Figure 1. The monthly demand for cannabis by quality.

specifically on conditional elasticities. As far as we are aware, there are no elasticity estimates (participation or conditional) for cannabis demand in South Africa. The illicit nature of cannabis use makes data collection difficult. By conducting an anonymous survey targeted at cannabis users, we were able to gather data on the cannabis market.

The Central Drug Authority (CDA) recognizes, in its National Drug Master Plan (NDMP) for 2013–2017, the necessity of studies on the dynamics of drug use, and has emphasized the need for evidence-based solutions (Republic of South Africa: Department of Social Development, Central Drug Authority, 2013). This study aims to address this need.

We find that medium-quality cannabis is almost twice as expensive as low-quality cannabis, while high-quality cannabis is nine times more expensive than low-quality cannabis. The price differential between low- and high-quality cannabis in South Africa is much larger than the price differential in the US (144%) (Davis et al., 2016) and Canada (20%) (Ouellet et al., 2017). Despite cannabis production being illegal in South Africa since 1929, the United Nations Office on Drugs and Crimes lists South Africa as one of the world's major cannabis-producing countries (United Nations Office on Drugs and Crime, 2006). Because cannabis grows abundantly in South Africa, low-quality cannabis (and even medium-quality cannabis) is cheap.

The price elasticity estimates fall very clearly in the inelastic range, indicating that an increase in the price of cannabis will result in a less than proportionate decrease in the quantity consumed. Depending on the model specification, we estimate that a 10% increase in the price per gram of cannabis is associated with an approximate 5–6% decrease in the quantity demanded. Our estimates are comparable to Ouellet et al.'s (2017) estimates of a conditional price elasticity of demand of between −0.42 and −0.60 in the Canadian population (Ouellet et al., 2017).

Similar to a recent Australian study (Clements, 2006), we find that an increase in the amount of cannabis in a typical transaction results in a decrease in price. Clements finds that doubling the package size decreases the unit price of cannabis by 25% (Clements, 2006). Our analysis shows that a doubling in package size will reduce the price by 30%. Clements links this decrease to risk: 'cutting' or 'splitting' cannabis into smaller package sizes implies a greater number of required transactions. The risk is conveyed via significant changes in its unit price (i.e. its discount elasticity) (Clements, 2006).

In terms of pricing, the cannabis market is fairly sophisticated, and acts as economic theory would predict. Other than the obvious finding that higher-quality cannabis is sold at a higher price than medium- and low-quality cannabis, the price increases for wealthier population groups, higher-income individuals, and cannabis users who live in wealthier provinces. This suggests that drug dealers have been quite successful in splitting the cannabis market into different market segments, and adjusting the price to what their customers can pay.

Although the international literature suggests that consumers of vice goods voluntarily ration their purchase quantities as a self-control strategy (Wertenbroch, 1998), we did not find this. In contrast, consumers in our sample bought both small and large amounts of cannabis (Table 2).

We note several limitations. Most importantly, our sample is not nationally representative, overrepresenting consumers in the Western Cape and Gauteng. The relatively small number of responses from African individuals skews the sample toward White and Colored South Africans. Since the data was collected from an online survey, the sample is also skewed toward those with internet access (and specifically those who are more active on Twitter and Facebook), and will thus over-represent individuals of higher socio-economic status.

It is likely that survey respondents are in favor of the legalization of cannabis, given that we approached some of the pro-legalization movements to publicize the survey. These consumers are likely to be intensive users and may under-represent infrequent users. In addition, these consumers might have a higher propensity to participate in research of this nature, especially since the cannabis legalization trial occurred in tandem with data collection.

A further limitation is that we did not consider the effect of alcohol prices on the demand for cannabis. Although Davis et al. (2016) included beer tax in their regressions to test whether alcohol is a substitute or a complementary good with respect to cannabis, the results were insignificant.

We focus on the conditional elasticity of demand, however, there may be some participation elasticity involved. For some people, the price may be near their threshold for participation. When asked to report their last purchase it may be

Table 5. OLS estimates of cannabis price.

	(1)	% (2)
Quality*population group (base: low*White)		
Low*African	−0.773***	−53.8
	(0.163)	
Low*Colored	−0.384***	−31.9
	(0.125)	
Low*Asian/Indian	−0.264*	−23.2
	(0.153)	
Medium*African	0.257***	29.3
	(0.097)	
Medium*Colored	0.306***	35.8
	(0.092)	
Medium*Asian/Indian	0.456***	57.8
	(0.127)	
Medium*White	0.633***	88.3
	(0.076)	
High*African	1.034***	181.2
	(0.139)	
High*Colored	1.415***	311.6
	(0.110)	
High*Asian/Indian	1.709***	452.3
	(0.132)	
High*White	1.640***	415.5
	(0.081)	
Ln (quantity purchased)	−0.436***	
	(0.015)	
Ln (income)	0.103***	
	(0.016)	
Gender (base: male)	–	
Female	−0.060*	−5.8
	(0.033)	
Age (base: 20–29)	–	
<20	−0.141***	−13.2
	(0.053)	
30–39	0.025	2.5
	(0.042)	
40–49	−0.092	−8.8
	(0.084)	
50+	−0.185**	−16.9
	(0.087)	
Province (base: Western Cape)	–	
Gauteng	0.058*	6.0
	(0.034)	
Other	−0.035	−3.4
	(0.038)	
Intensity (base: less than daily)	–	
Daily	0.068**	7.0
	(0.031)	
Constant	2.508***	
	(0.153)	
Observations	1831	
R-squared	0.680	

Dependent variable: logged price per gram. Robust standard errors in parentheses.
***$p < .01$, **$p < .05$, *$p < .1$.

that they considered making another purchase since their last reported purchase, but decided not to because the price was too high. This would not be an issue with the more intensive users (daily or near-daily), which make up the bulk of our sample, but may be apparent in the less intensive users.

Although we wanted to test for endogeneity in the price variable, the lack of variables in our dataset did not allow for this. Davis et al. (2016) used instrumental variables (IV) (electricity price, distance to Mexico City, and medical cannabis designation) as a companion to ordinary least square (OLS) models and found that the OLS and IV estimates are similar, providing some assurance that endogeneity of the price variable is weak in the demand equation.

Since the level of quality was left to the respondent to interpret in the survey instrument, the observations on quality may be correlated with respondent attributes, which may result in biased estimates.

Conclusion

Legalizing cannabis is likely to impact significantly both the demand for and supply of cannabis. It is unclear at this point how a legalized market will be regulated, if at all. If the government wishes to impose an excise tax on the product, as has occurred in some states in the US, an understanding of the market dynamics is important. The aim of this study was to investigate the demand for cannabis among existing users.

Our single most important finding is that the consumption of cannabis is significantly affected by price but that the price elasticity falls in the inelastic range. The market for cannabis, even though it operates in the shadow economy, is subject to the same economic forces as other consumer products. The price responsiveness of cannabis is similar to that of cigarettes. Should cannabis be legalized, the government could use some of its experience about the regulation, and especially taxation, of cigarettes as a model for cannabis.

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Note

1. An average exchange rate for 2017 ($1 = R13.30) is used.

ORCID

Nicole Vellios http://orcid.org/0000-0003-1488-0179
Corné van Walbeek http://orcid.org/0000-0003-0933-435X

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